



Project Laboratory 1

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1 Abstract

This document presents an investigation into discrepancy learning methods for nonlinear dynamical systems using Sparse Identification of Nonlinear Dynamics (SINDy) and Neural Network based correction models. Several systems were examined, including a linear spring-mass-damper system, a falling ball with nonlinear aerodynamic drag, and the nonlinear Van der Pol oscillator.

The objective of the work was to:

1. Reconstruct governing differential equations from simulated measurement data with SINDy to understand it
2. Investigate the effect of measurement noise and numerical differentiation
3. Apply sparse regression techniques to identify compact physical models
4. Introduce artificial discrepancies into physical systems
5. Learn and compensate these discrepancies using both SINDy and Neural Networks

The results demonstrate that SINDy can successfully recover interpretable governing equations, while discrepancy learning methods estimate and compensate for the difference between the nominal model and the observed system dynamics.